

Where You Swim in Toronto Matters More Than Yesterday's Test*

Two beaches exceed the *E. coli* limit eight times as often as the cleanest, while a warning based on the previous day's samples misses most days over the limit

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September 25, 2026

Sentence 1: the general area (Toronto's beaches and the daily *E. coli* warning). Sentence 2: data and method (98,370 samples at ten beaches, 2007 to 2026, from Open Data Toronto; days over the limit, a same-day model, yesterday against today). Sentence 3: headline result (Marie Curtis Park East and Sunnyside over the limit on about a third of days, eight times Gibraltar Point; yesterday's samples catch 37% of days over the limit). Sentence 4: implication (knowing which beach you are at tells a swimmer more than yesterday's reading).

1 Introduction

*Paragraph 1, context: Toronto's ten swimming beaches are sampled daily in summer; the City posts a warning when *E. coli** exceeds 100 per 100 mL. Swimmers read the posting as today's water.**

E. coli is a sign of faecal pollution. It lives in the gut of people and other warm-blooded animals, so finding it in lake water means that sewage, storm-water runoff or bird droppings have arrived recently, bringing the viruses and bacteria that make swimmers ill [1]. Swimming in water above Toronto's standard raises the risk of ear, eye, nose, throat and skin infections, and swallowed water can cause stomach illness. Young children, older people and people with weakened immune systems are most at risk [2].

Paragraph 2, what the paper does: 20 years of results from Open Data Toronto; two questions, whether some beaches go over the limit more often than others, and how well one day's result predicts the next. Gap: the City publishes the results but not a comparison across beaches or a check of the one-day delay.

*Code and data are available at: <https://github.com/seandata8/toronto-beaches-ecoli>.

Paragraph 3, what was found (specific numbers): Marie Curtis Park East 34% and Sunnyside 32% of days against Gibraltar Point 4%; same ranking in 19 of 20 years; a warning based on yesterday's samples catches 37% of days over the limit, and 37% of warnings are needed.

Paragraph 4, why it matters: choose the beach, not the day; the posting is a weak guide.

Paragraph 5, structure: Section 2 describes the data and how a sample becomes a number, Section 3 the same-day model, Section 4 the results, Section 5 what they mean and their limits. The appendices cover cleaning and the check on simulated data.

2 Data

2.1 Source and context

The data are the *E. coli* results from Toronto's beach water sampling program, published by Toronto Public Health on the City's open data portal [3,4]. The program is run jointly by Toronto Public Health, Toronto Water, the police Marine Unit and the City's parks division, under a provincial protocol whose stated aim is to reduce water-borne illness [3]. Every summer, from Victoria Day to Labour Day, staff take a water sample each day at four to six fixed sites along the shore of each of the city's ten supervised beaches (Figure 1). A laboratory counts the *E. coli* in each sample, and when the level at a beach is too high, Toronto Public Health posts signs warning against swimming [5].

The data set was downloaded on 22 September 2026. It has 102,534 rows, one for each site on each sampling day from 3 June 2007 to 16 September 2026, and each row gives the beach, a site code, the date the sample was collected, the *E. coli* result and the site's location. After leaving out rows with no result, one implausible reading, two sites listed under the wrong beach and dates outside the season (Appendix A), 98,370 results remain, from 50 sites on 2,019 days across 20 years.

Toronto's standard is 100 *E. coli* per 100 mL of water; Ontario and Health Canada use 200 [1,2]. This paper uses 100, the level at which Toronto posts warnings.

The data describe water, not people, and contain no personal information. The stakes lie in the advice built on them: a needless warning costs swimmers a beach day, and a missed one exposes them to illness [6]. The City also offers daily swimming conditions for download [5], but this paper uses the open data because they give every result, so each day's status can be worked out the same way for all twenty years.

The analysis uses Python [7] with Polars [8], NumPy [9], statsmodels [10], pointblank [11], matplotlib [12] and contextily [13], and the paper is written in Quarto [14].

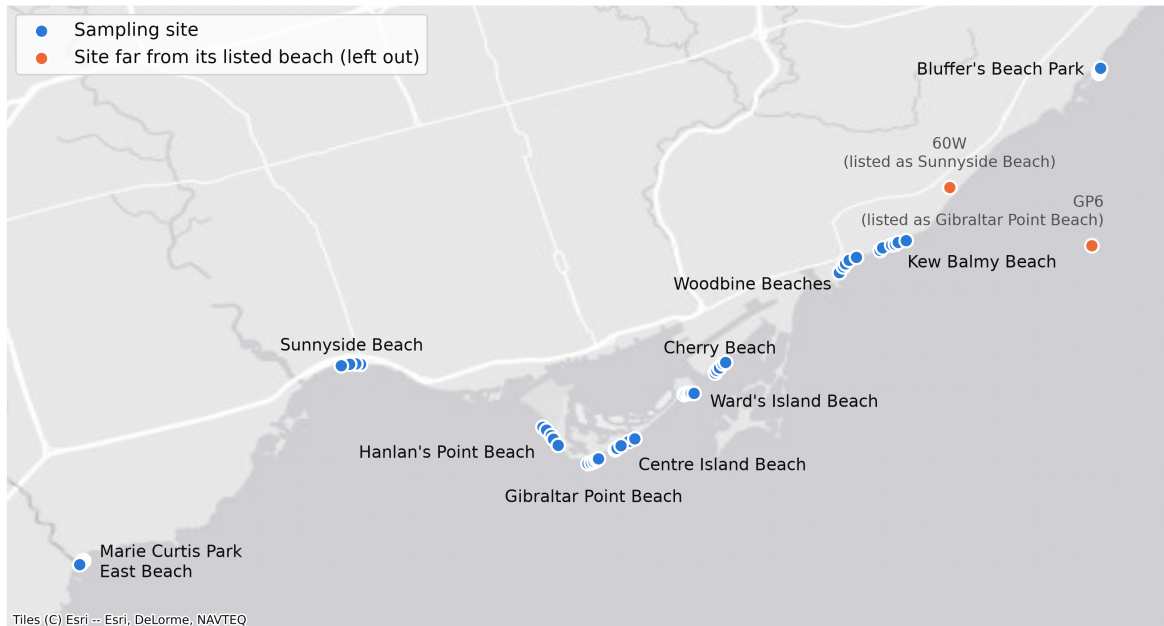


Figure 1. Each of Toronto's ten swimming beaches is sampled at four to six sites along the shore. Blue points are the 50 sites this paper uses. The two orange points, sites 60W and GP6, were added in 2026 and sit 14 to 16 km from the beaches they are listed under, so they are left out. Map tiles from Esri.

2.2 Measurement

A number in the dataset is the end of a chain that starts with a person wading into the lake. At each site, staff fill a sterile bottle to 200 mL, plunging it to about 30 cm below the surface where people swim. The provincial protocol asks for at least five samples per beach, spread across the swimming area [15]. Most Toronto beaches have five sites; Kew Balmly has six and Sunnyside four. The bottles are kept cold and dark, and must reach a Public Health Ontario laboratory and be tested within a day of collection [15].

The laboratory uses membrane filtration [15]. The water is passed through a filter with pores small enough to trap bacteria, and the filter is laid on a growth medium on which *E. coli* colonies can be told apart from other bacteria. Overnight, each *E. coli* cell grows into a colony large enough to see; the colonies are counted and the count is scaled to *E. coli* per 100 mL. The method detects more than 90% of *E. coli* strains, but cells stressed by the lake may not grow and other bacteria can interfere, so a single result carries error [15]. The process takes about a day [2,5], so a result reaches the public a day after the water it describes. When no sample can be taken, usually because of bad weather, the row is kept with its result left blank [2]; 3.5% of rows are blank.

The way results are reported leaves its mark on the numbers, as Figure 2 shows. Nearly all results (99.9%) are multiples of 10, and half (49%) are exactly 10. Ten is the lowest value the laboratory reports, so a result of 10 means “10 or fewer”: the data have a floor. From 2018 a second limit appears at the top. Results of exactly 1,000 become common while results above 1,000 become rare, and in 2026 none is above 1,000, although before 2018 readings reached 15,820. From 2018, then, a result of 1,000 is best read as “1,000 or more”. Both limits come from the reporting, not the water. The floor hides differences among the clean beaches, and the ceiling hides how bad the worst days were.

2.3 *E. coli* results by beach

Table 1 summarises the results at each beach, and Figure 3 shows every one of them. Results span more than three orders of magnitude, so the figures use a logarithmic scale, on which each equal step is a tenfold increase: 10, 100, 1,000. Most results are low. The median, the middle result when all are put in order, is 20, and at eight of the ten beaches it is 10 or 20. Marie Curtis Park East and Sunnyside stand apart. Their medians are 50 and 60, only about a quarter and a fifth of their results are at the floor, and about a third are above 100. At Gibraltar Point, the cleanest beach, 61% of results are at the floor and 6% are above 100.

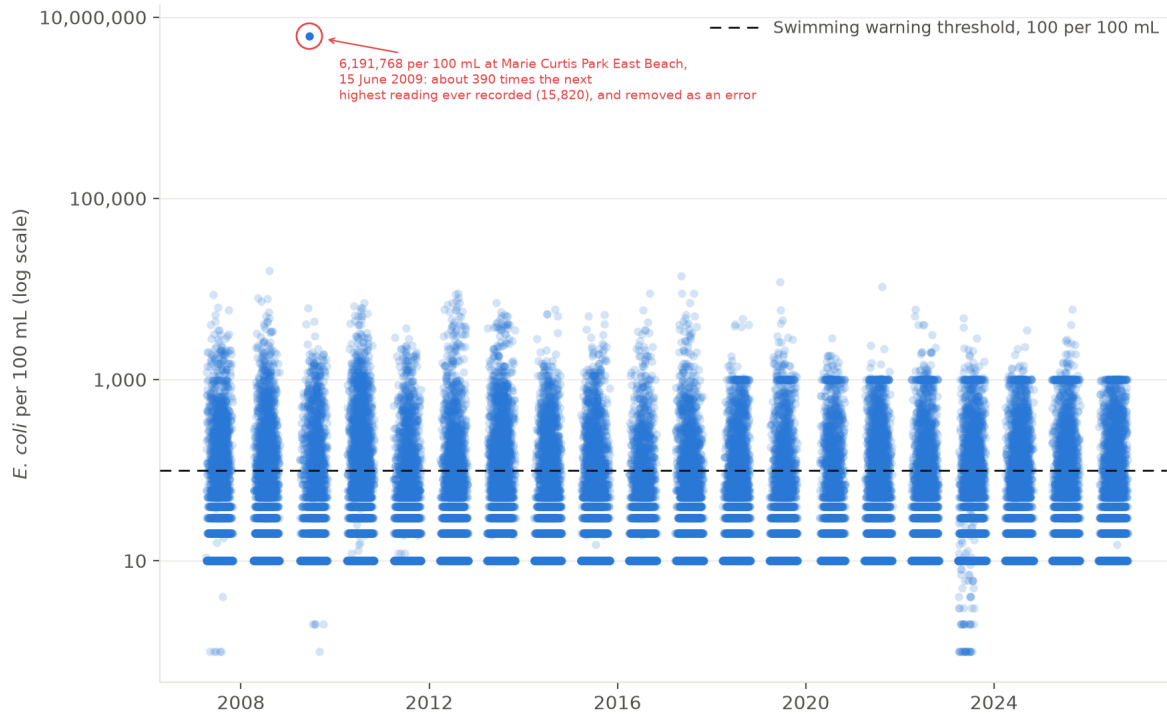


Figure 2. Every result the City has published, 2007 to 2026. One point is one water sample (98,991 in total), before any cleaning. Points are spread sideways within each year so that they can be seen; the *E. coli* count itself is never moved. Most results are reported in multiples of 10, which is why the low readings fall into separate rows. The laboratory reports no results as zero, and “no colonies detected” is assumed to be reported as 10. From 2018 on, many results are reported as exactly 1,000, and these are assumed to be readings of 1,000 or more. The dashed line marks 100 *E. coli* per 100 mL, the level at which the City posts a swimming warning. The ringed point is the only reading the analysis removes.

Table 1. Results at each beach, 2007 to 2026. A result is the number of *E. coli* per 100 mL in one water sample. Ten is the lowest value the laboratory reports, and 100 is Toronto’s limit for swimming. The 90th percentile is the level that one result in ten exceeds. Beach-days are days with at least four results. Beaches run from the largest share of results above 100 to the smallest.

Beach	Sites	Results	Me- dian	90th per- centile	At 10	Above 100	Beach- days
Marie Curtis Park East Beach	5	9,612	50	590	27%	35%	1,925
Sunnyside Beach	4	8,006	60	400	20%	34%	1,986
Centre Island Beach	5	9,645	20	220	40%	19%	1,929
Kew Balmy Beach	6	12,047	20	250	47%	19%	2,011
Bluffer’s Beach Park	5	10,006	10	150	55%	13%	2,001
Woodbine Beaches	5	10,035	10	110	59%	11%	2,005
Ward’s Island Beach	5	9,757	10	110	58%	10%	1,952
Cherry Beach	5	10,012	10	90	58%	9%	2,003
Hanlan’s Point Beach	5	9,689	10	80	61%	8%	1,939
Gibraltar Point Beach	5	9,561	10	60	61%	6%	1,911

2.4 Days over the limit

Toronto’s limit applies not to a single result but to the geometric mean of a beach’s results. A geometric mean of n results multiplies them together and takes the n th root; put another way, it averages the logarithms of the results and converts that average back. It suits counts that span orders of magnitude because a single extreme result does not dominate it. Five results of 10, 10, 10, 10 and 1,000 have an ordinary average of 208 but a geometric mean of 25.

In this paper a beach-day is over the limit when the geometric mean of that day’s results is above 100. Only days with at least four results count, four rather than five so that Sunnyside, with four sites, is included. That gives 19,662 beach-days, of which 2,816 (14%) are over the limit. Figure 4 shows every one. On 22% of beach-days every result was at the floor of 10.

A day over the limit is not the same as a day the beach was posted, for three reasons. The data do not record whether a warning went up. A study of the program describes posting decisions based on a geometric mean over two days rather than one [6]. And because results take a day, a warning posted today reflects yesterday’s water. “Over the limit” here means that the water sampled that day exceeded 100, judged after the fact.

Counting days rather than results changes the shares only a little. At every beach the share of days over the limit is 1.2 to 5.5 percentage points below the share of results above 100: Marie Curtis Park East, for example, was over the limit on 34% of days and had 35% of results above

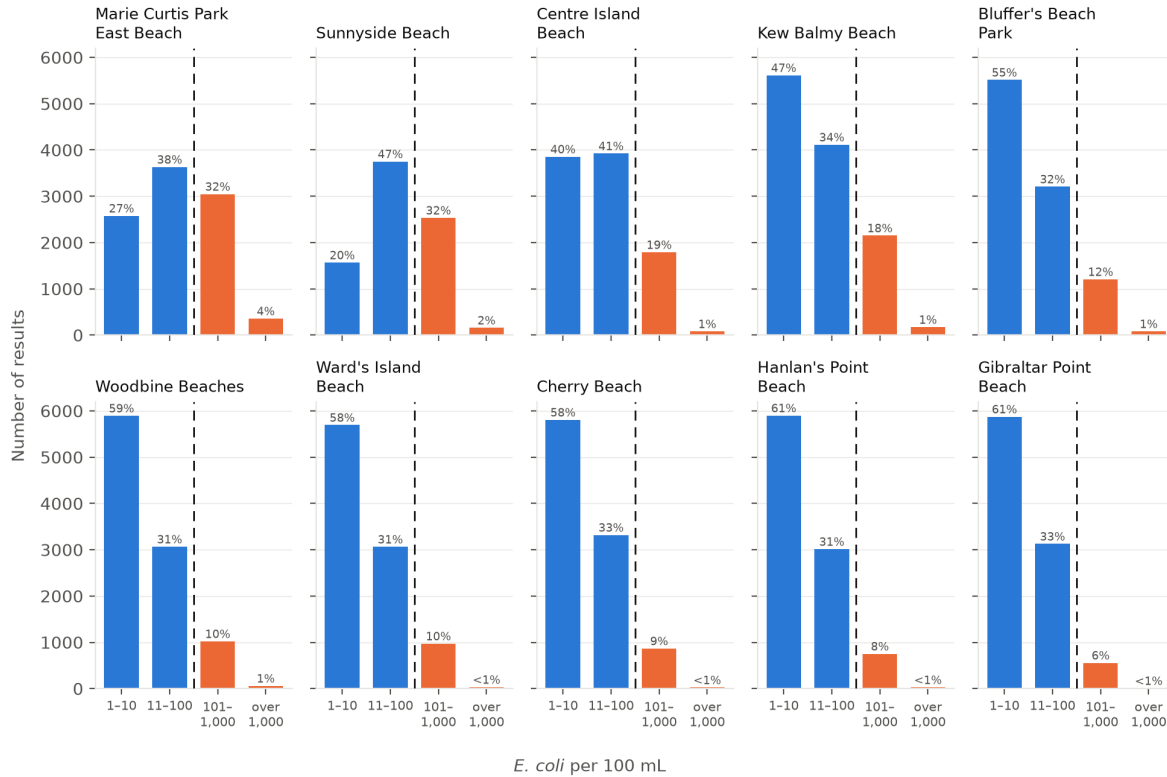


Figure 3. How high the results get, by beach. Every result the analysis uses (98,370), with each bar labelled by its share of that beach's results. Cleaning leaves out the reading of 6,191,768 removed as an error, two sites added in 2026 that sit far from the beaches they are listed under, and dates outside the sampling season. The orange bars, right of the dashed line, are the results above 100 *E. coli* per 100 mL, the limit the City of Toronto sets for safe swimming. Panels run from the beach with the largest share above that limit to the smallest. Each bar counts single samples rather than days: a beach is judged on the geometric mean of its four to six samples, and by that measure Marie Curtis Park East Beach was over the limit on 657 of its 1,925 sampled days (34%), against 35% of its samples.

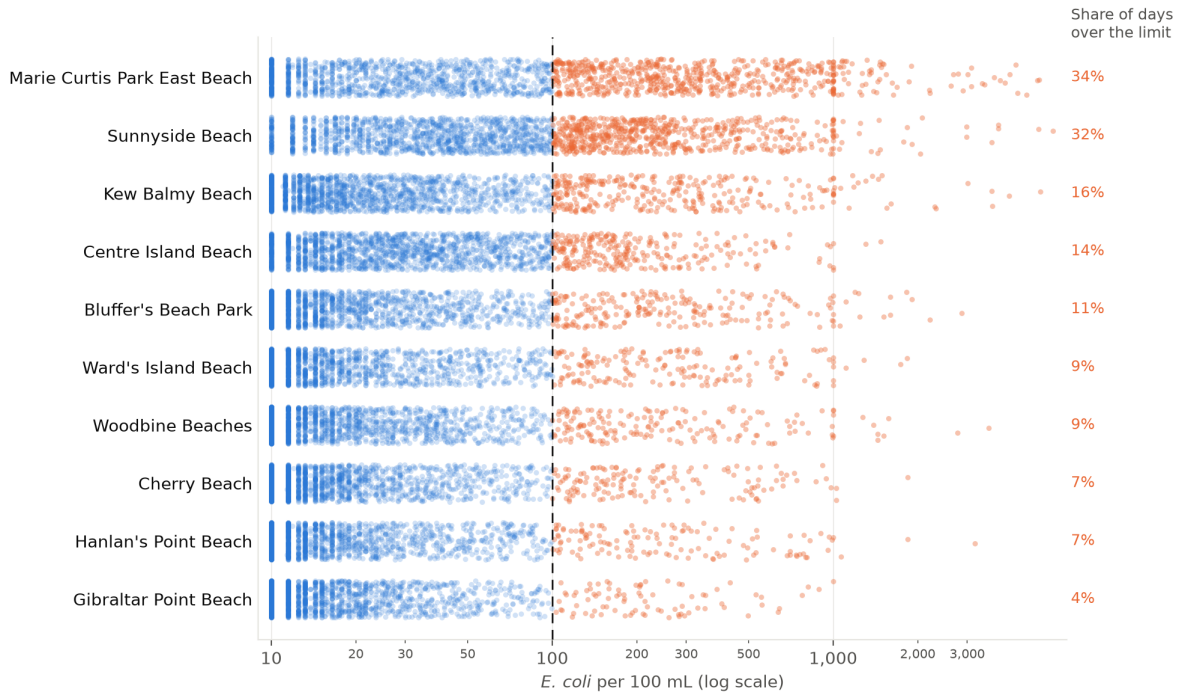


Figure 4. Every beach-day, by beach. One point is the geometric mean of a beach's samples on one day, the figure the City's limit applies to (19,662 days in total). Days with fewer than four results are left out. Points are spread vertically so that they can be counted; the value itself is never moved. Orange points, right of the dashed line, are the days above 100 *E. coli* per 100 mL. Half of all samples are reported at 10, the lowest the laboratory measures, so the column at 10 is the days on which every sample was at that floor, and the stripes just above it are the values a geometric mean of four to six multiples of ten can take. The 17 days whose mean falls below 10 are off the left of the axis.

100. Averaging five results might be expected to pull most days back under the limit, but it does not, because a beach's sites tend to rise and fall together. When one is high, the others usually are too.

The ceiling matters more once results are averaged. Since 2018, 34 beach-days include a result of exactly 1,000 yet have a geometric mean below 100. On about 8 of them a true reading of 2,000 or less would have put the day over the limit, so the count of days over the limit from 2018 on is, if anything, slightly low.

3 Model

Points: the question the model answers (holding the day's weather fixed, does one beach run higher than another?). Equation: $\log_{10}$ daily geometric mean on beach and date, every symbol explained in words; the date term compares each beach only with the others sampled that day, so rain is not a confounder. What a coefficient means (0.3 on the $\log_{10}$ scale is about twice as high). Days over the limit come in runs, so the standard errors treat days in the same calendar month as related (about twice as wide as treating each day as independent). 45 pairs, Holm's correction explained. The floor and ceiling shrink the gaps to about 65% of their true size in simulation (Appendix B).

4 Results

4.1 Two beaches are over the limit far more often than the rest

Points: shares of days over the limit (34.1%, 31.8% down to 4.4%). Not a few bad years: same top two in 19 of 20 years, exception 2021. Not the ceiling: split at 2018 barely moves the shares. Model: Marie Curtis and Sunnyside 0.34 above the daily average (more than twice as high), not different from each other, different from every other beach; 31 or 32 of 45 pairs differ. Centre Island ranks above Kew Balmy in the model but below it in share of days, and why.

Figure 5 shows the model estimates.

4.2 Yesterday's samples are a weak guide to today's water

Points: 18,998 pairs of consecutive days at the same beach. A warning based on yesterday catches 37% of days over the limit, and 37% of such warnings are needed. 14% of days are over the limit overall, 37% after a warning, 10% after an all-clear (about 3.5 times as likely). Leave out overall accuracy.

Table (to add): `?@tbl-warning`, two-by-two from `other/results/warning-confusion-matrix.csv`.

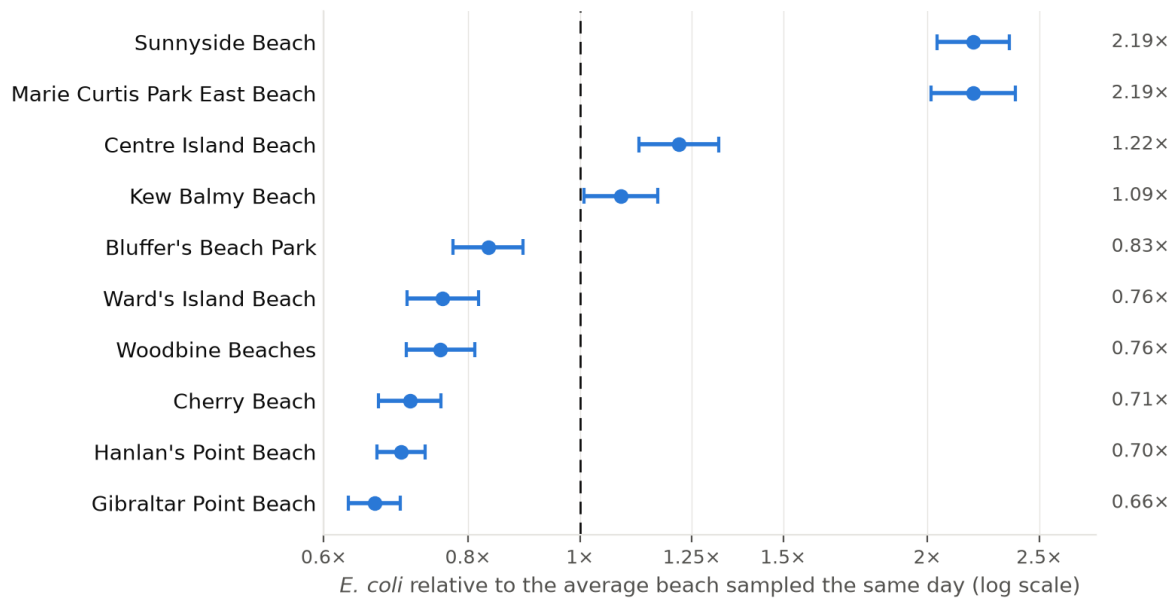


Figure 5. Placeholder caption: each beach's typical level against the average beach on the same day, on the log10 scale, with 95% intervals. The gaps are understated: the reporting floor and ceiling shrink them to about two thirds of their true size.

5 Discussion

5.1 Which beach you choose matters more than which day you go

Points: summary of the finding; possible reasons from geography and literature (Etobicoke Creek at Marie Curtis, the Humber River near Sunnyside, storm-water outfalls, island against mainland beaches). Hypotheses, not tested here.

5.2 A warning describes yesterday's water

Points: the one-day lag; water changes faster than the testing can follow; same-day methods such as qPCR (literature).

5.3 Weaknesses

Points: days over the limit are computed, not recorded postings; the floor and ceiling understate the gaps; no weather data; Sunnyside's four sites add noise (effect under one percentage point in simulation).

5.4 Next steps

Points: add rainfall and wind to predict days over the limit. Compare days over the limit with the warnings actually posted: the City's "Beach Condition Data for Researchers" download (JSON) may record them.

Appendix

A Data cleaning

Points: what was removed and why (the reading of 6,191,768, sites 60W and GP6 with a footnote on the City's reply and the download date of 22 September 2026, dates outside the season, days with fewer than four results). Tests run on the cleaned data.

B Checking the method on simulated data

Points: simulated data with known beach differences, the floor and ceiling; the analysis recovers the ordering; censoring leaves about 65% of a true gap; the site-count effect.

C Pairwise comparisons

Points: the 45 pairs with Holm-corrected p-values, and the check with partial months merged. Table from `other/results/beach-pairwise-differences.csv` and `beach-pairwise-grouping-check.csv`.

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